EC-390: Digital Signal Processing

LECTURE NO 4: *Classifications of Discrete Time Signals*

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Organization of Lecture

1. Classifications of Discrete Time Signals.
   i. Deterministic and Nondeterministic Signals
   ii. Periodic and Aperiodic Signals.
   iii. Symmetric and Asymmetric Signals.
   iv. Energy Signals and Power Signals.
   v. Causal and Noncausal Signals.

1. Classifications of Discrete Signals

i- <u>Deterministic and Non-deterministic signals:</u>
The signals that can be completely specified by mathematical equations are called <u>deterministic signals</u>. The step, ramp, exponential and sinusoidal signals are deterministic signals.
The signals whose characteristics are random in nature are called <u>nondeterministic signals</u>. The noise signals from various sources are best examples of nondeterministic signals.

ii- <u>Periodic and Aperiodic Signals:</u>
- For the discrete time signal, the condition of periodicity is:

$$x(n) = x(n+N)$$

> where N = fundamental Period

- When 'N' is the fundamental, the periodic signals will also satisfy the condition:

$$x(n+kN) = x(n)$$

> where 'k' is an integer

- Periodic signals are power signals.
- The sinusoidal and complex exponential signals are periodic signals when fundamental frequency f_0 = rational

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Problems Related to Periodic & Aperiodic Signals:

Example 1: Determine whether the given signals are periodic or Aperiodic.

a) $x(n) = \sin(5\pi n)$

Soln

$x'(n) = \sin(2\pi f_0 n)$

Compare $x(n)$ & $x'(n)$ we get:

$2\cancel{\pi} f_0 \cancel{n} = 5\cancel{\pi}\cancel{n}$

$f_0 = \frac{5}{2} = \frac{\text{integer}}{\text{integer}}$ = rational number

When fundamental frequency 'f_0' is a rational number, the signal is said to be periodic

As $f_0 = \frac{5}{2} \longrightarrow$ Fundamental Period

Therefore $N = 2$.

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b) $x(n) = \cos(4n)$

$\underset{\sim}{\text{sol}^n}$

$x'(n) = \cos(2\pi f_0 n)$

Compare $x(n)$ & $x'(n)$ we get:

$2\pi f_0 \cancel{n} = 4\cancel{n}$

$f_0 = \frac{4}{2\pi} = \frac{\text{integer}}{\text{Non-integer}}$ = irrational = Aperiodic Signal

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c) $x(n) = \cos\left(\frac{11}{10}\pi n\right) + \sin\left(\frac{7}{10}\pi n\right)$

Sol^n:

The above signal can be written as:

$x(n) = x_1(n) + x_2(n)$

Considering $x_1(n) = \cos\left(\frac{11}{10}\pi n\right)$

$x_1'(n) = \cos(2\pi f_0 n)$

comparing $x_1(n)$ & $x_1'(n)$ we get:

$2\cancel{\pi} f_0 \cancel{n} = \frac{11}{10}\cancel{\pi}\cancel{n}$

$f_0 = \frac{11}{10\times 2} = \frac{11}{20} =$ periodic with $N_1 = 20$

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Considering $x_2(n) = \sin\left(\frac{7}{10}\pi n\right)$

$$x_2'(n) = \sin(2\pi f_0 n)$$

Comparing $x_2(n)$ & $x_2'(n)$ we get:

$$2\cancel{\pi} f_0 \cancel{n} = \frac{7}{10} \cancel{\pi} \cancel{n}$$

$$f_0 = \frac{7}{20} = \text{periodic with } N_2 = 20$$

Since '$\frac{N_1}{N_2}$' is a ratio of two integers, the given sequence '$x(n)$' is periodic with fundamental period $N = 20 \times 1$ ('which is LCM of N_1 & N_2)

$$\begin{array}{r|l} 20 & 20, 20 \\ \hline 1 & 1, 1 \\ \hline \end{array}$$

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d) $\sin\left(\frac{5\pi}{2}n\right) \cdot \cos\left(\frac{2\pi}{3}n\right) = x(n)$

Sol^n:

the above signal can be written as:

$$x(n) = x_1(n) \cdot x_2(n)$$

Considering $x_1(n) = \sin\left(\frac{5\pi}{2}n\right)$

$$x_1'(n) = \sin(2\pi f_0 n)$$

while comparing we get:

$$2\cancel{\pi} f_0 \cancel{n} = \frac{5\cancel{\pi}}{2}\cancel{n}$$

$$f_0 = \frac{5}{4} = \frac{\text{int}}{\text{int}} = \text{periodic with } N_1 = 4$$

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Considering $x_2(n) = \cos\left(\frac{2\pi}{3} n\right)$

$$x_2'(n) = \cos(2\pi f_0 n)$$

While comparing we get:

$$2\cancel{\pi} f_0 \cancel{n} = \frac{2\cancel{\pi}}{3} \cancel{n}$$

$$f_0 = \frac{2}{6} = \frac{1}{3} = \frac{int}{int} = \text{periodic with } N_2 = 3$$

Fundamental Period $N = N_1 \times N_2 = 4 \times 3 = 12$

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e) $x(n) = 1 + e^{j\frac{2\pi}{3}n} - e^{j\frac{4\pi}{7}n}$

Soln

Above signal can be written as:

$$x(n) = x_1(n) + x_2(n) - x_3(n)$$

consider $x_1(n) = 1 \Rightarrow$ DC signal with arbitrary Period $= 1$ i.e., $N_1 = 1$

Considering $x_2(n) = e^{j\frac{2\pi}{7}n}$

$$x_2'(n) = e^{j2\pi f_0 n}$$

When comparing we get:

$$2\pi f_0 n = \frac{2\pi}{3} n$$

$$f_0 = \frac{1}{3} = \text{periodic with } N_2 = 3$$

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Considering $x_3(n) = e^{j\frac{4\pi}{7}n}$.

$x_3'(n) = e^{j2\pi f_0 n}$

when comparing we get:

$$2\pi f_0 \cancel{n} = \frac{4\pi}{7} \cancel{n}$$

$$f_0 = \frac{\cancel{4}2}{\cancel{14}7} = \frac{2}{7} = \text{periodic with } N_3 = 7$$

The fundamental period 'N' of given sequence x(n) will be:

$N =$ LCM of N_1, N_2 & N_3

1	1, 3, 7
3	1, 3, 7
7	1, 1, 7
	1, 1, 1

$N = 1 \times 3 \times 7$

$N = 21$

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iii) Symmetric (Even) and Asymmetric (odd) signals:

* Any signal $x(n)$ can be represented as sum of even and odd components. That is

$$x(n) = x_e(n) + x_o(n)$$

where

$$x_e(n) = \text{even compenent(s) of } x(n)$$
$$x_o(n) = \text{odd component(s) of } x(n)$$

Even signal:

A D.T signal $x(n)$ is said to be an even (symmetric) signal if it satisfies the condition:

$$x(n) = x(-n)$$

Odd signal:

A D.T signal $x(n)$ is said to be an odd (Asymmetric) signal if it saties the condition:

$$x(-n) = -x(n)$$

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Even part of the signal can be written as:

$$x_e(n) = \frac{1}{2}\left[x(n) + x(-n)\right]$$

Odd part of the signal can be written as:

$$x_o(n) = \frac{1}{2}\left[x(n) - x(-n)\right]$$

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Problems related Even and Odd Signals:

Example 1: Determine the even and odd parts of the signal:

$x(n) = 3^n$

Soln:

for even signal: $x(n) = x(-n) = 3^{-n}$

for odd signal: $x(-n) = -x(-n) = -3^{-n}$

Even part:

$$x_e(n) = \frac{1}{2}\left[3^{+n} + 3^{-n}\right]$$

Odd part:

$$x_o(n) = \frac{1}{2}\left[3^{n} - 3^{-n}\right]$$

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Example 2: Determine the even and odd parts of the signal or sequence.

$x(n) = \{-3, 1, \underline{2}, -4, 2\}$

Sol^n:

for even signal : $x(n) = x(-n) = \{2, -4, \underset{\uparrow}{2}, 1, -3\}$

for odd signal : $x(-n) = -x(-n) = \{-2, 4, \underset{\uparrow}{-2}, -1, 3\}$

Even part:

$x_e(n) = \frac{1}{2}\left[\{-3, 1, 2, -4, 2\} + \{2, -4, 2, 1, -3\}\right.$

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$$= \frac{1}{2}\left[(-3+2), (1-4), (2+2), (-4+1), (2-3)\right]$$

$$x_e(n) = \{-0.5, -1.5, \underset{\uparrow}{2}, -1.5, -0.5\}$$

Odd Part:

$$x_o(n) = \frac{1}{2}\left[\{-3, 1, 2, -4, 2\} + \{-2, 4, -2, -1, 3\}\right]$$

$$x_o(n) = \{2.5, 2.5, \underset{\uparrow}{0}, 2.5, 2.5\}$$

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(iv) Energy and Power Signals:

* Energy E of a D.T $x(n)$ is defined as:

$$\text{Energy} = E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

* If energy of a D.T signal is finite and non-zero. The signal is said to energy signal

* In energy signal, the energy of signal $x(n)$ is finite and average power is zero.

$$0 < E < \infty \quad \text{and} \quad P_{avg} = 0$$

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* The average power of a D.T signal $x(n)$ is defined as:

$$\text{Power} = P = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$

* If a signal has a finite average power and non zero as well. The signal is said to be a power signal.
* The average power of signal is finite and energy is infinite.

$$0 < P_{avg} < \infty \text{ and } E = \infty$$

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Problems Related to Energy and Power Signals:

Example 1: Determine the Energy and Power of a unit sample signal. i.e.,

$$\delta(n) = \begin{cases} 1 & \text{for } n = 0 \\ 0 & \text{for } n \neq 0 \end{cases}$$

Sol^n:

$$\text{Energy} = E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$= \sum_{n=0}^{0} |1|^2$$

$$\boxed{E = 1 \text{ Joules}}$$

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$$\text{Power} = P = \underset{N\to\infty}{\text{Limit}} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$

$$= \underset{N\to\infty}{\text{Limit}} \frac{1}{2N+1} \left[\cancel{\sum_{n=-N}^{-1} |x(n)|^2}^{\,0} + \sum_{n=0}^{0} |x(n)|^2 \right]$$

$$= \underset{N\to\infty}{\text{Limit}} \frac{1}{2N+1} \left[0 + 1 \right]$$

$$= \lim_{N\to\infty} \frac{1}{2N+1} \cdot (1)$$

Applying Limit $N\to\infty$

$$= \frac{1}{2(\infty)+1} = \frac{1}{\infty} = 0$$

$$\boxed{P_{avg} = 0}$$

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Example 2: Determine the energy and Power of a unit Ramp sequence.

$$x(n) = \begin{cases} n; & n \geq 0 \\ 0; & n < 0 \end{cases} \Rightarrow \boxed{\text{HOME WORK}}$$

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Example 3: Determine whether the given signal is energy signal or power signal.

$$x(n) = \left(\frac{1}{4}\right)^n u(n) = 0.25\, u(n)$$

Sol^n:

Here $x(n) = \left(\frac{1}{4}\right)^n u(n)$ for all n

$$= \left(\frac{1}{4}\right)^n \; ; n \geq 0$$

$$x(n) = (0.25)^n ; n \geq 0$$

$$\text{Energy} = E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$= \sum_{n=-\infty}^{\infty} |0.25^n|^2$$

$$= \cancel{\sum_{n=-\infty}^{-1} |0.25^n|^2} + \sum_{n=0}^{\infty} |0.25^n|^2$$

$$= \sum_{n=0}^{\infty} |0.25^2|^n \quad \text{—— (1)}$$

According to infinite geometric series formula.

1. Classifications of Discrete Signals

$$\boxed{\sum_{n=0}^{\infty} c^n = \frac{1}{1-c}}$$ → Infinite Geometric series Sum formula

So applying it on eq ① we get:

$$\text{Energy} = E = \sum_{n=0}^{\infty} |0.25^2|^n = \frac{1}{1-0.25^2} = \frac{1}{1-0.0625}$$

$$\boxed{E = 1.067 \text{ Joules}}$$

$$\text{Power} = P_{avg} = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$

$$P_{avg} = \lim_{N\to\infty} \frac{1}{2N+1} \left[\sum_{n=-N}^{-1} |x(n)|^2 + \sum_{n=0}^{N} |x(n)|^2 \right]$$

$$= \lim_{N\to\infty} \frac{1}{2N+1} \left[\underbrace{\sum_{n=-N}^{-1} |0.25^n|^2}_{0} + \sum_{n=0}^{N} |0.25^n|^2 \right]$$

$$P_{avg} = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=0}^{N} |0.25^2|^n \quad \text{— ②}$$

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According to finite geometric series sum formula

$$\boxed{\sum_{n=0}^{N} C^{n} = \frac{C^{N+1} - 1}{C - 1}} \Rightarrow \text{Finite Geometric series sum formula}$$

Apply it on eq ② we get:

$$P_{avg} = \lim_{N \to \infty} \frac{1}{2N+1} \left[\frac{0.0625^{N+1} - 1}{0.0625 - 1} \right]$$

Applying Limit $N \to \infty$ we get

$$\boxed{\begin{array}{l} -\infty + 1 = \infty \\ - A^{\infty} = 0 \\ - 2(\infty) = \infty \end{array}}$$

$$= \frac{1}{2(\infty)+1} \left[\frac{0.0625^{\infty+1} - 1}{-0.9375} \right]$$

$$= \frac{1}{\infty+1} \left[\frac{0.0625^{\infty} - 1}{-0.9375} \right] = \frac{1}{\infty} \left[\frac{0 - 1}{-0.9375} \right]$$

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$$= \frac{1}{\infty}[1.0667]$$

$$= 0 \times [1.0667]$$

$$P_{avg} = 0 \text{ Watt}$$

$1/\infty = 0$

Comments!

Energy is finite i.e., $E = 1.067$ Joules and $P_{avg} = 0$. Therefore the given signal is an energy signal.

Example 4: Determine whether the given signal is energy signal or power signal.

$x(n) = \sin(\pi/3)n$ (HOME WORK)

1. Classifications of Discrete Signals

Causual and Non causal Signals

* A signal is said to be causal, if it is defined for $n \geq 0$. Therefore if $x(n)$ is causal, then $x(n) = 0$ for $n < 0$.
* A signal is said to be non-causal, if it is defined for either $n \leq 0$, or for both $n \leq 0$ and $n > 0$

Examples of Causal and Noncausal Signals

$$x(n) = \{\underset{\uparrow}{1}, -1, 2, -2, 3, -3\}$$
$$x(n) = \{\underset{\uparrow}{2}, 2, 3, 3, \ldots\ldots\}$$

Causal Signals

$$x(n) = \{1, -1, 2, -2, 3, \underset{\uparrow}{-3}\}$$
$$x(n) = \{\ldots\ldots, 2, 2, 3, \underset{\uparrow}{3}\}$$

Non-Causal Signals

References: Textbooks

1. *"Discrete-Time Signal Processing", International Edition, by Oppenheim, Alan V; Schafer, Ronald W, (2013), Pearson.*
2. *"Digital Signal Processing Using Matlab", 4th Edition by Vinay K. Ingle, John G. Proakis, (2016), Cengage Learning.*
3. *Related material from open source, mainly internet.*

Appendix

"USEFUL FORMULAS USED IN ENERGY AND POWER SIGNALS"

i) $\sum_{n=0}^{N} C^n = \frac{C^{N+1}-1}{C-1}$.

ii) $\sum_{n=0}^{\infty} C^n = \frac{1}{1-C}$.

iii) $\sum_{n=0}^{N} 1^n = N+1$

iv) $\sum_{n=-N}^{N} 1^n = 2N+1$

v) $\sum_{n=-N}^{N} |x(n)|^2 = \sum_{n=-N}^{-1} |x(n)|^2 + \sum_{n=0}^{N} |x(n)|^2$

vi) $\infty + 1 = \infty$

vii) $\infty - 1 = \infty$

viii) $\frac{\infty}{\text{anything}} = \infty$

ix) $A^{(\infty)} = 0$ $(0 < A < 1)$; when A is b/w 0 & 1

x) $\frac{\text{anything}}{\infty} = 0$

xi) $\frac{\infty}{\infty}$ = undefined we must avoid it.

xii) $e^{j\pi} = \cos(\pi) + j\sin(\pi) = -1$ [Euler's identity in terms of sine & cosine]

or $e^{j\pi}+1=0$

xiii) $\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$

xiv) $\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$

xv) $\sin^2\theta = \frac{1-\cos 2\theta}{2}$

xvi) $\cos^2\theta = \frac{1+\cos 2\theta}{2}$ → power reducing formula.

xvii) $\sum_{n=-\infty}^{-1} C^n = \sum_{n=1}^{\infty} C^{-n} = \sum_{n=0}^{\infty} C^{-n} - 1$